

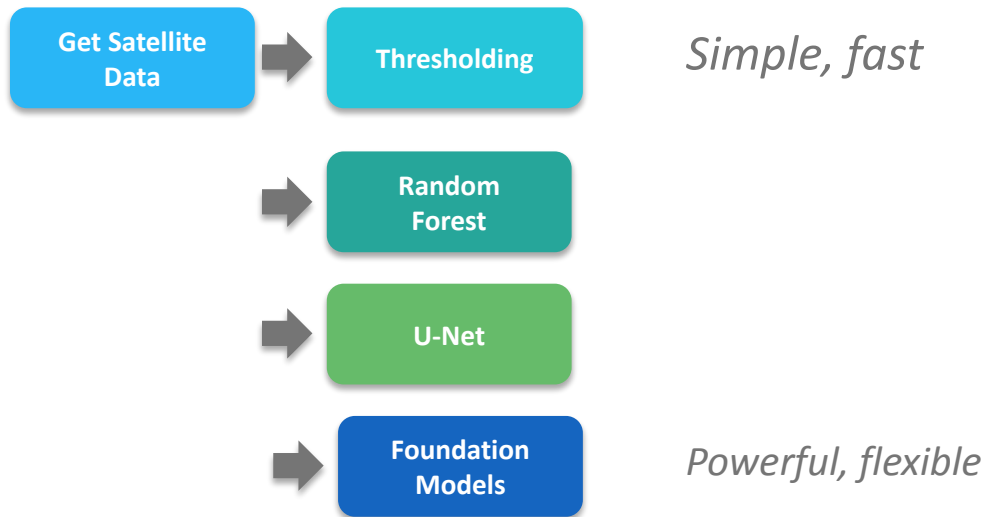
Monitoring Water from Space

An introduction to AI

Dr. Sindhura | CTO, Smart Bhujal | 2026

Talk Roadmap

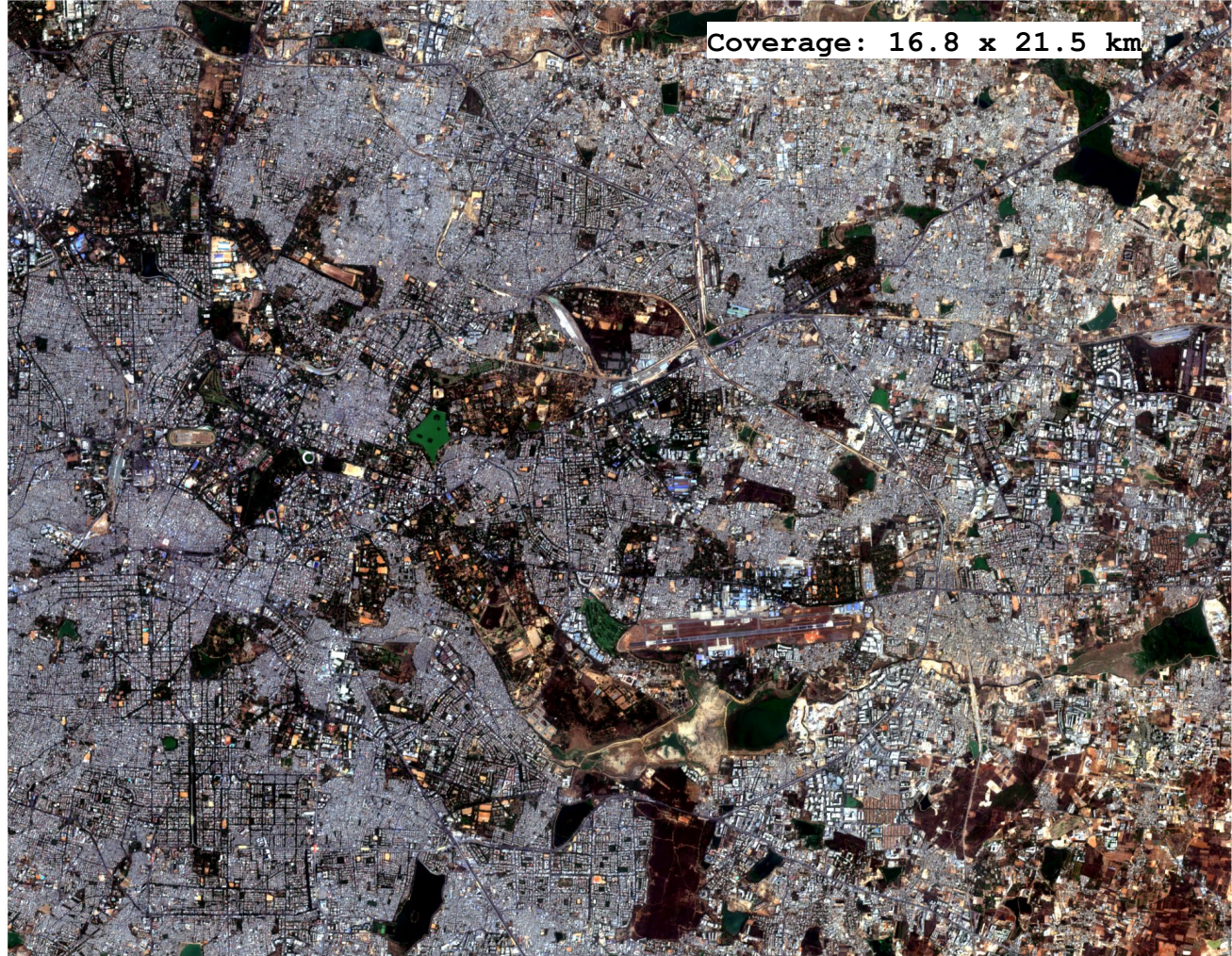
To monitor/track water bodies through satellite imagery using AI



We'll progressively use better techniques — from simple image processing to AI

Coverage: 16.8 x 21.5 km

**Bangalore has lost
79% of its lakes in
the last 4 decades.
*Can we track this ?***



The Problem

Manual lake surveys are slow, expensive, and infrequent

Encroachment happens between survey cycles

Flood-prone zones change as the city grows

"What if you could monitor every water body, every 5 days, for free?"

Data

Sentinel-2 — Free Eyes in the Sky

ESA satellite, launched 2015

13 spectral bands — visible, NIR, SWIR

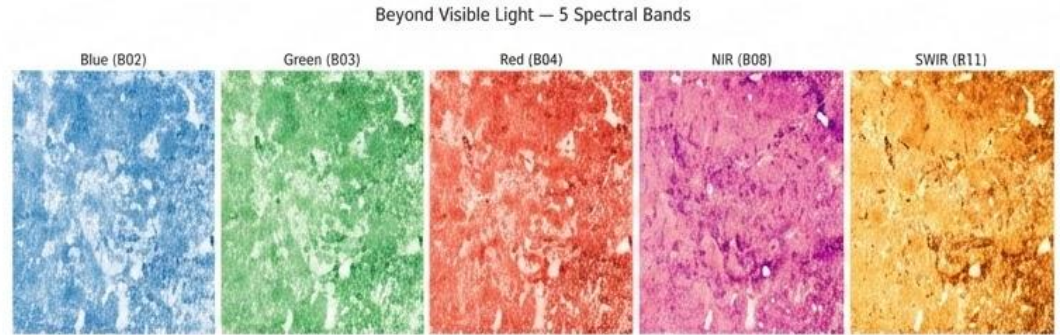
10m resolution — individual buildings visible

5-day revisit — near real-time monitoring

Free and open data — no cost, no account needed

Sentinel-2: Free Eyes in the Sky & Beyond Visible Light

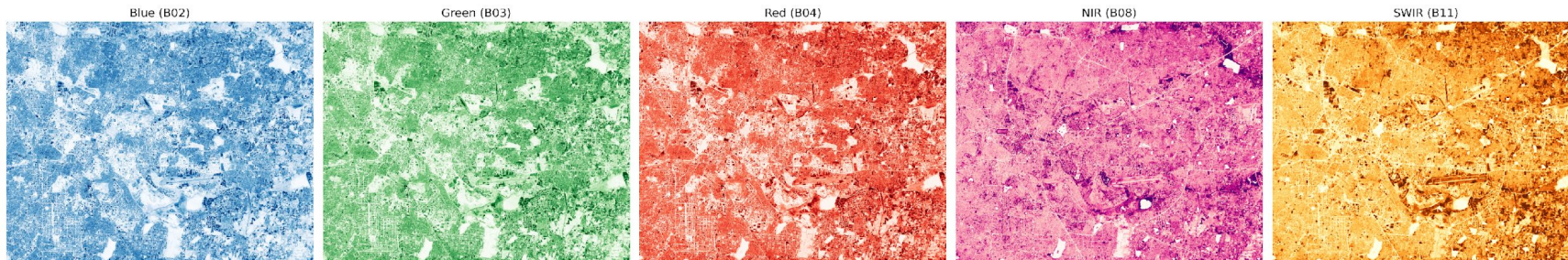
- ESA satellite, launched 2015
- 13 spectral bands — visible, NIR, SWIR
- 10m resolution — individual buildings visible
- 5-day revisit — near real-time monitoring
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NIR (B08): Absorbed by water, reflected by vegetation | SWIR (B11): Absorbed by water, reflected by buildings

Beyond Visible Light — Why It Matters

Beyond Visible Light — 5 Spectral Bands



NIR (B08): Absorbed by water, reflected by vegetation | SWIR (B11): Absorbed by water, reflected by buildings

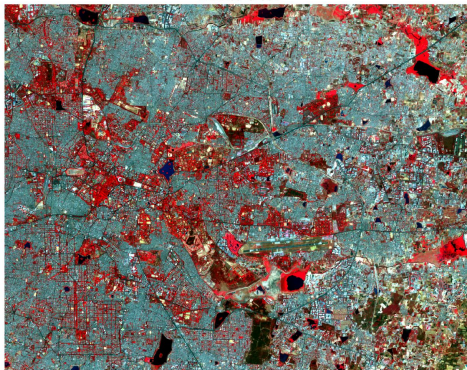
Band Composites — Seeing the Invisible

Band Composites — Seeing the Invisible

True Color (RGB)



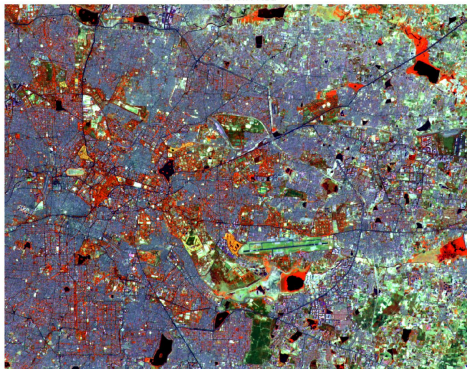
NIR-Red-Green



SWIR-NIR-Red



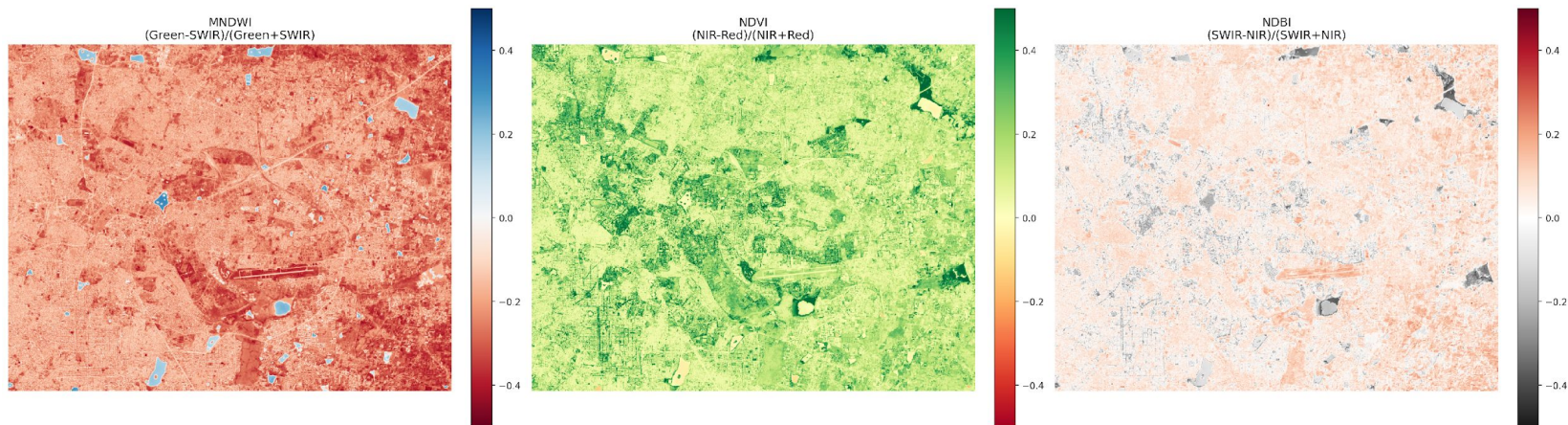
NIR-SWIR-Red



Three Spectral Indices

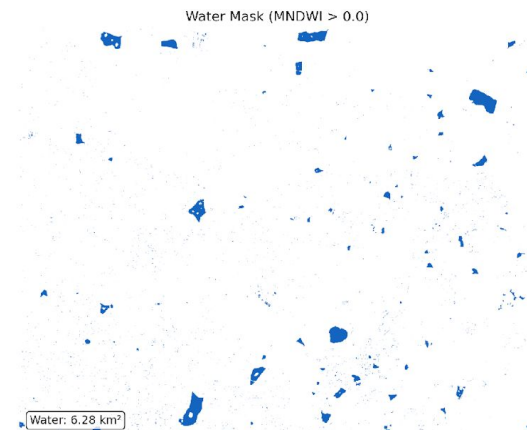
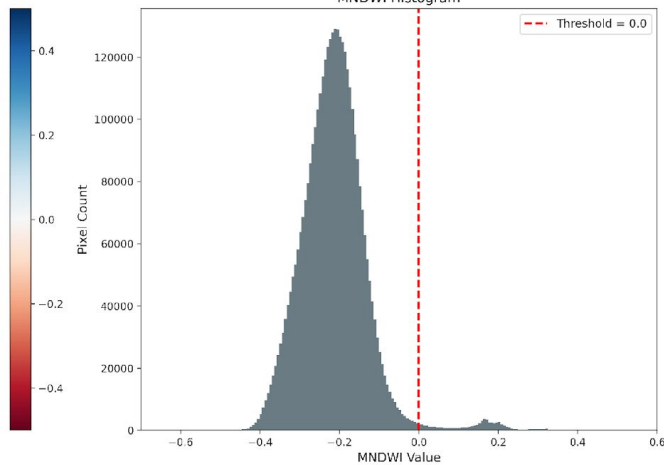
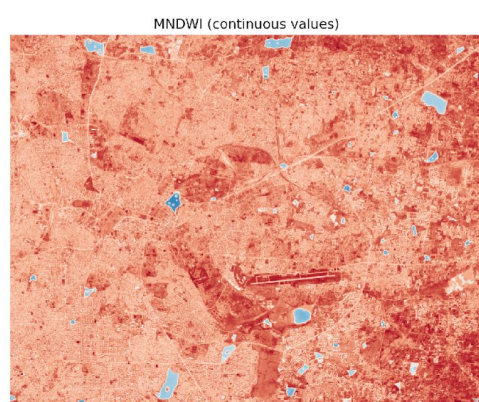
Index	Formula	Detects
MNDWI	$(\text{Green} - \text{SWIR}) / (\text{Green} + \text{SWIR})$	Water (urban)
NDVI	$(\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$	Vegetation health
NDBI	$(\text{SWIR} - \text{NIR}) / (\text{SWIR} + \text{NIR})$	Built-up surfaces

Three Spectral Indices — One Number per Pixel



What is Thresholding? — "Like Setting a Pass Mark"

What is Thresholding? — Pick a Cutoff, Classify Above/Below



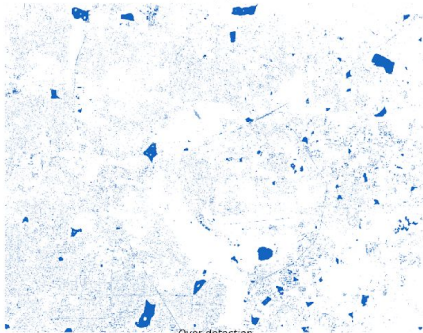
Threshold Sensitivity

Threshold Sensitivity — Choosing the Right Cutoff Matters

Threshold = -0.1



Water: 20.85 km²



Over-detection
(shadows, wet soil)

Threshold = 0.0



Water: 6.28 km²



Good balance

Threshold = 0.1



Water: 4.50 km²



Under-detection
(misses shallow)

Threshold = 0.2

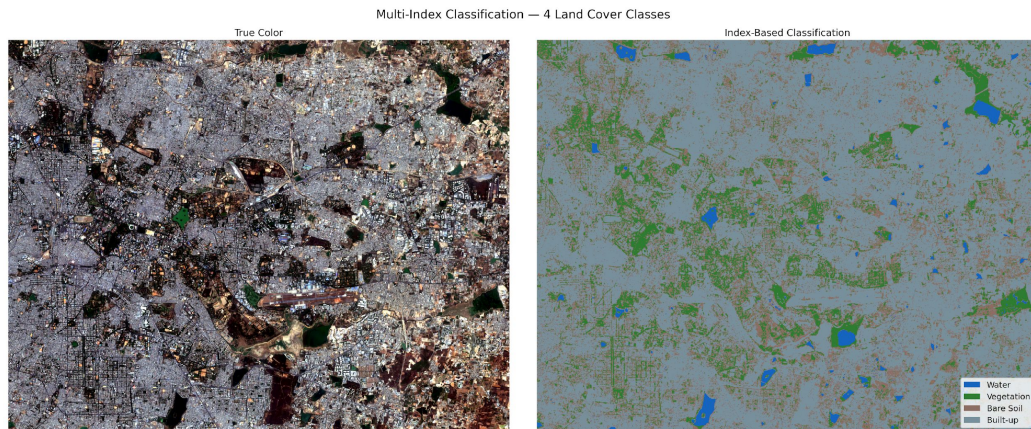


Water: 1.32 km²



Only deep water

Cleanup + Multi-Index Classification



Why Machine Learning?

Thresholds fail on:

- Building shadows misclassified as water
- Turbid/polluted water missed by MNDWI
- Single fixed thresholds don't adapt across seasons
- No spatial context (pixel-by-pixel, independently)

"Thresholds are rules **WE** write. **ML** learns rules **FROM** data."

Thresholding:

WE write the rules

WE choose the thresholds

Rules are simple (if/else)

Machine Learning:

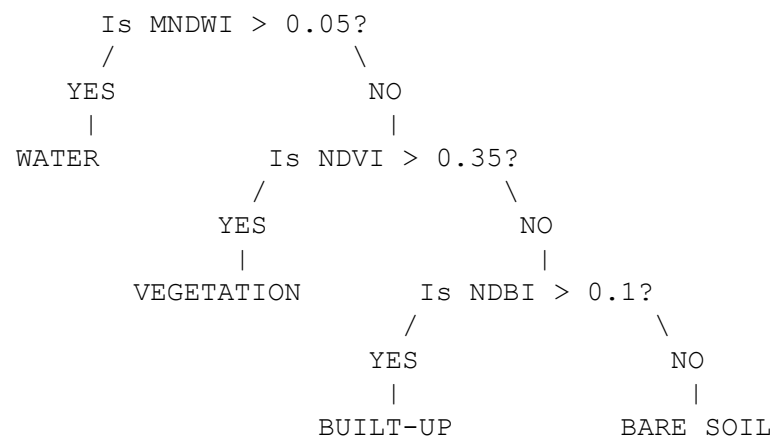
WE provide examples

THE ALGORITHM learns the rules

Rules can be complex and non-linear

Decision Trees and Random Forest

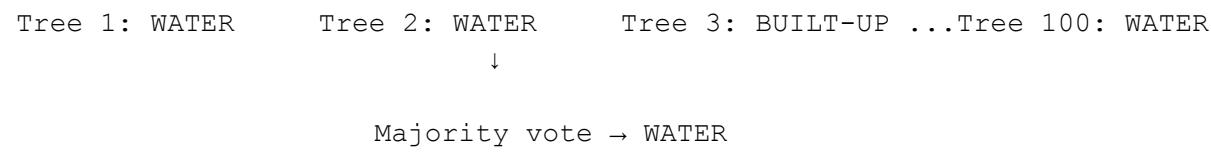
A **Decision Tree** asks a series of yes/no questions about the features:



A single tree is fragile — change the data slightly and the tree changes completely.

A **Random Forest** builds **100 trees**, each trained on a random subset of data.

Each tree votes, and the **majority wins**.



Decision Tree: Yes/no questions splitting data

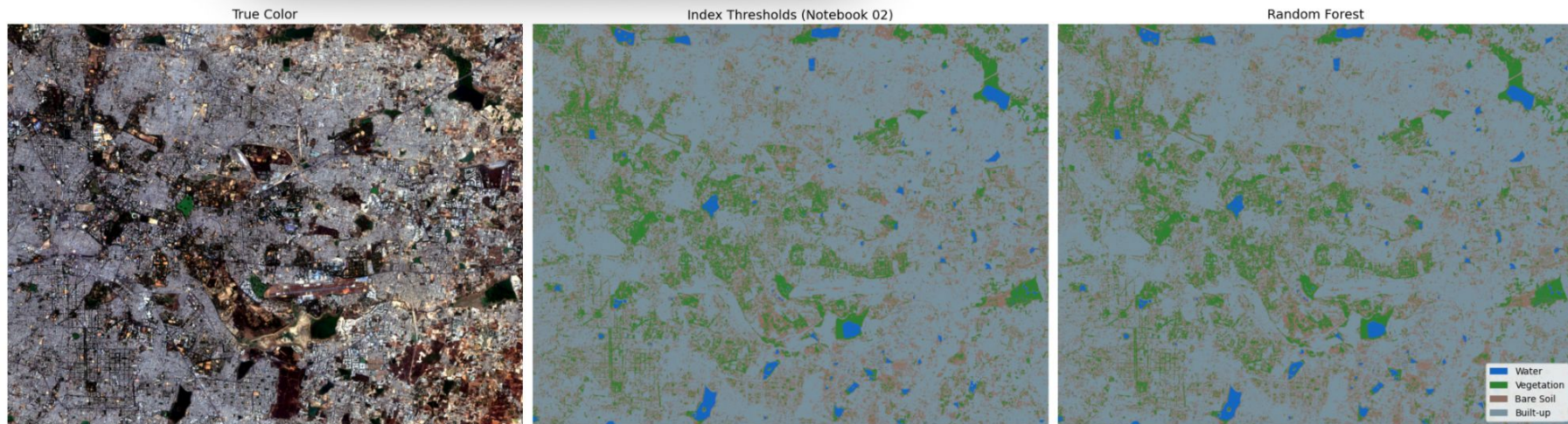
"Is MNDWI > 0?" → "Is NDVI < 0.3?" → Classify

Random Forest: 100 trees, each sees different data

Majority vote wins — errors cancel out

"Like asking 100 engineers instead of one"

Each pixel = 8 features: 5 bands + 3 spectral indices

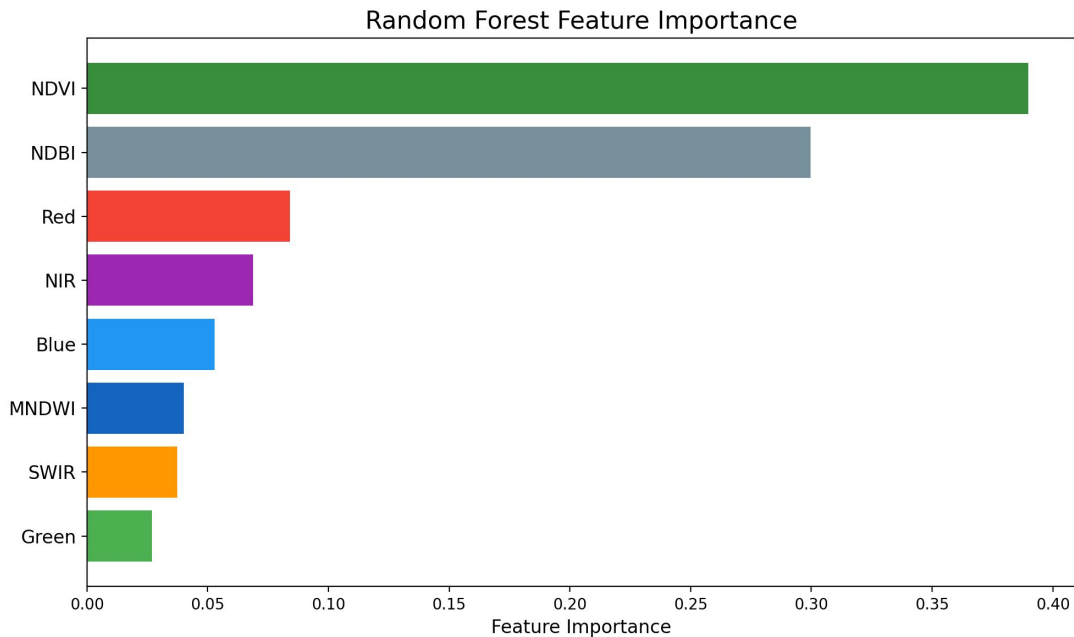


Limitation of Random Forest: No spatial awareness¶

Look closely at the Random Forest output — it has a **salt-and-pepper** appearance. Individual pixels are misclassified because Random Forest treats each pixel **independently**. It doesn't know that a pixel is surrounded by water pixels.

Imagine labeling a map one pixel at a time vs. looking at the whole neighborhood. That's the difference between Random Forest and deep learning.

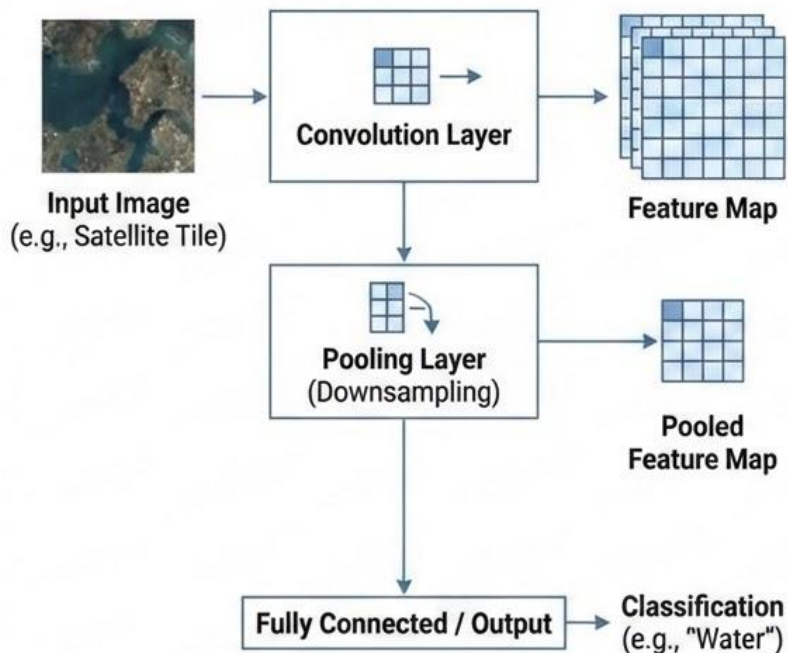
What the Model Relies On



MNDWI is the top feature — the model confirms what water engineers already know

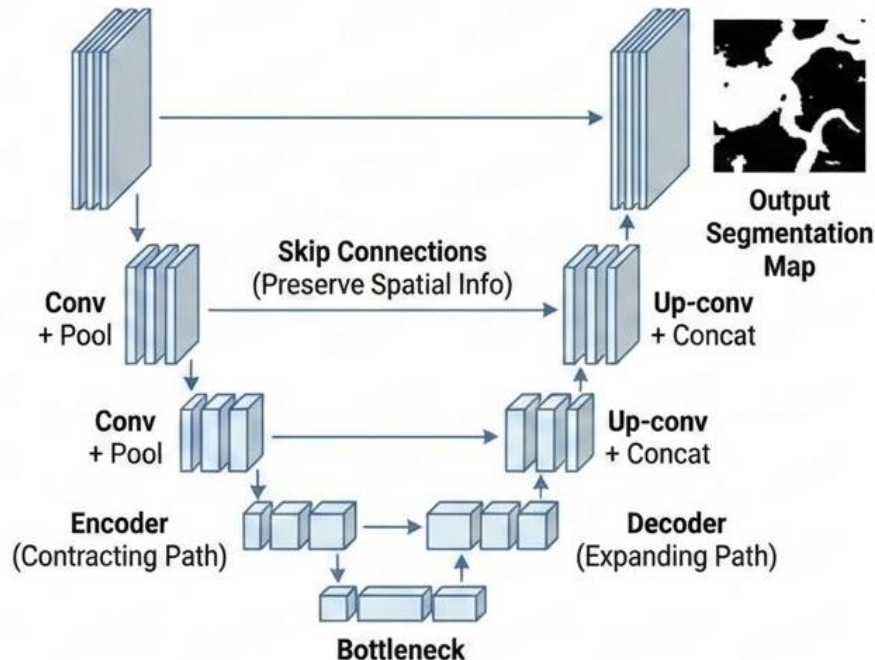
Deep Learning Architectures: CNN Basics & U-Net

CNN Basics (Feature Extraction)



Key Ops: Convolution (filters), Pooling (resizing).
Learns spatial hierarchies of features.

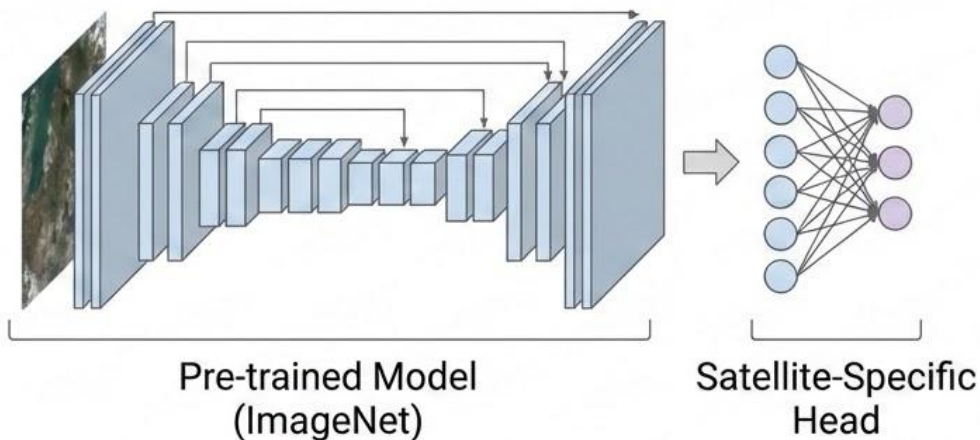
U-Net Architecture (Semantic Segmentation)



Encoder-Decoder structure with Skip Connections.
Classifies every pixel (e.g., Water Mask).

Transfer Learning

Reuse a model pre-trained on 1.2M photos (ImageNet)

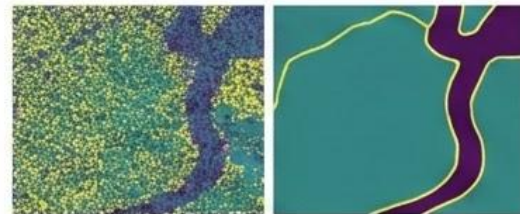


≡ **What transfers:** edges, textures, spatial patterns

🏗️ **What we add:** satellite-specific classification head

“You don’t need millions of satellite training samples”

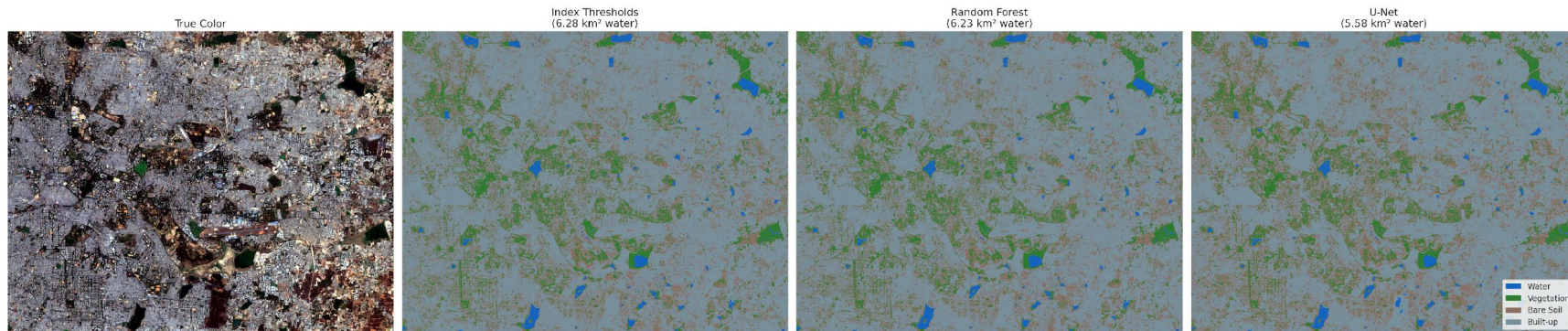
Result: U-Net produces smoother boundaries, fewer salt-and-pepper misclassifications than Random Forest



Random Forest U-Net with Transfer Learning

Classification Comparison

Classification Comparison — Thresholds vs ML



Water area: Thresholds 6.28 km² | Random Forest 6.23 km² | U-Net 5.58 km²

For regulatory work, choose the method that matches your accuracy needs

Foundation Models — The New Paradigm

"Hiring an experienced engineer vs. training a fresh graduate"

Pre-trained on millions of images with no labels (self-supervised)

Fine-tune for your task with very little labeled data

Model	Trained On	Parameters	Use Case
DINOv2 (Meta)	142M natural images	21M-1.1B	General vision
Prithvi-EO (IBM+NASA)	4.2M satellite images	300M-600M	Earth observation
Clay (Dev Seed)	Global Sentinel-2	300M	Geospatial

All Methods — Grand Comparison

Land Use Classification — Bangalore — All Methods



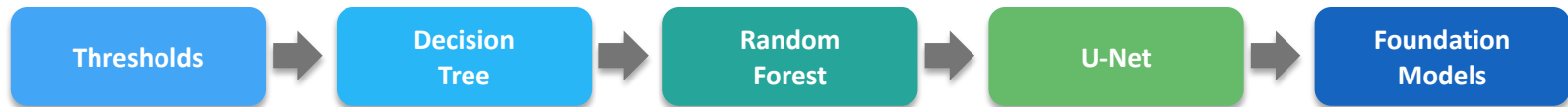
Water Area Detection — All Methods

Method	Water Area Detected
Index Thresholds	6.28 km ²
Random Forest	6.23 km ²
U-Net	5.58 km ²
DINOv2 + Spectral	6.13 km ²

When to Use What — Decision Guide

Situation	Method	Why
Quick field check	MNDWI threshold	Fast, no training needed
Regular monitoring	Random Forest	Fast, interpretable, 8 features
Precise boundaries	U-Net	Spatial context, smooth edges
New area, few labels	Foundation model	Pre-trained knowledge transfers

The ML Progression



Simple, fast

Powerful, flexible

"Start simple. Add complexity only when you need it."

Real Applications

Lake monitoring: Track surface area changes over seasons/years using MNDWI time series

Flood mapping: Pre/post-event MNDWI comparison to measure inundation extent in km²

Watershed health: NDVI trends reveal vegetation cover changes affecting runoff

Urban drainage: NDBI maps impervious surfaces for runoff coefficient estimation

Thank You

Questions?

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